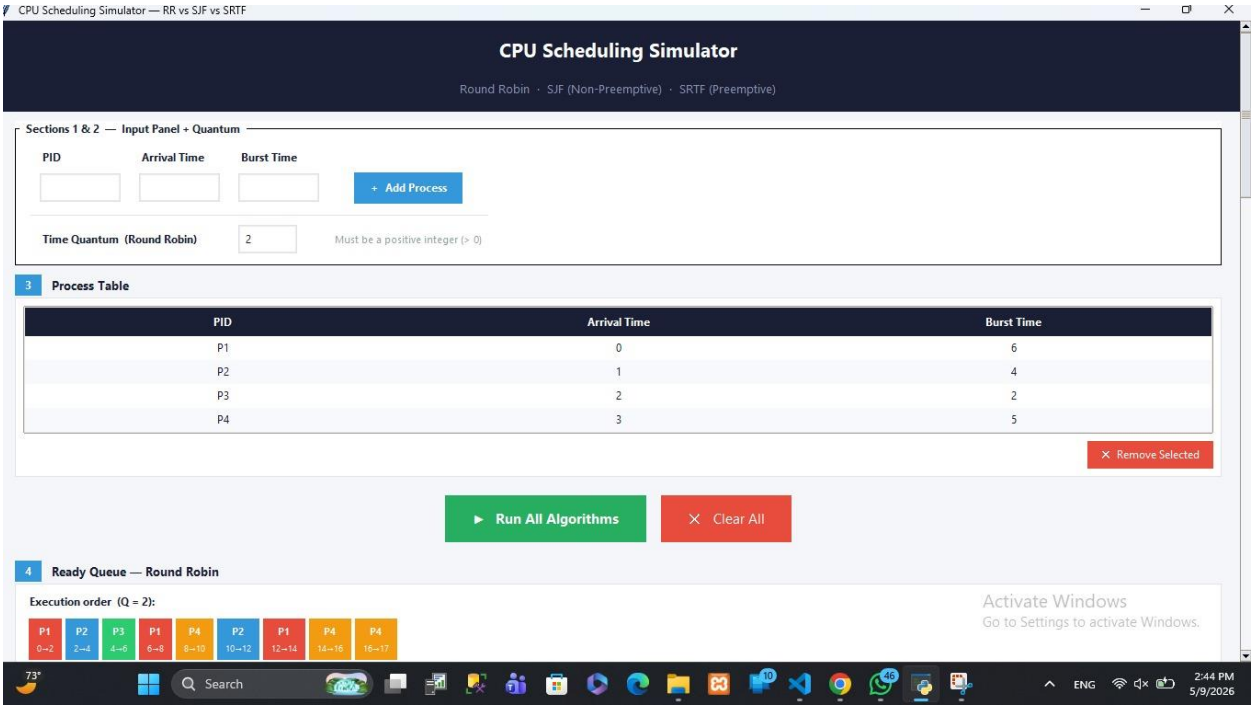
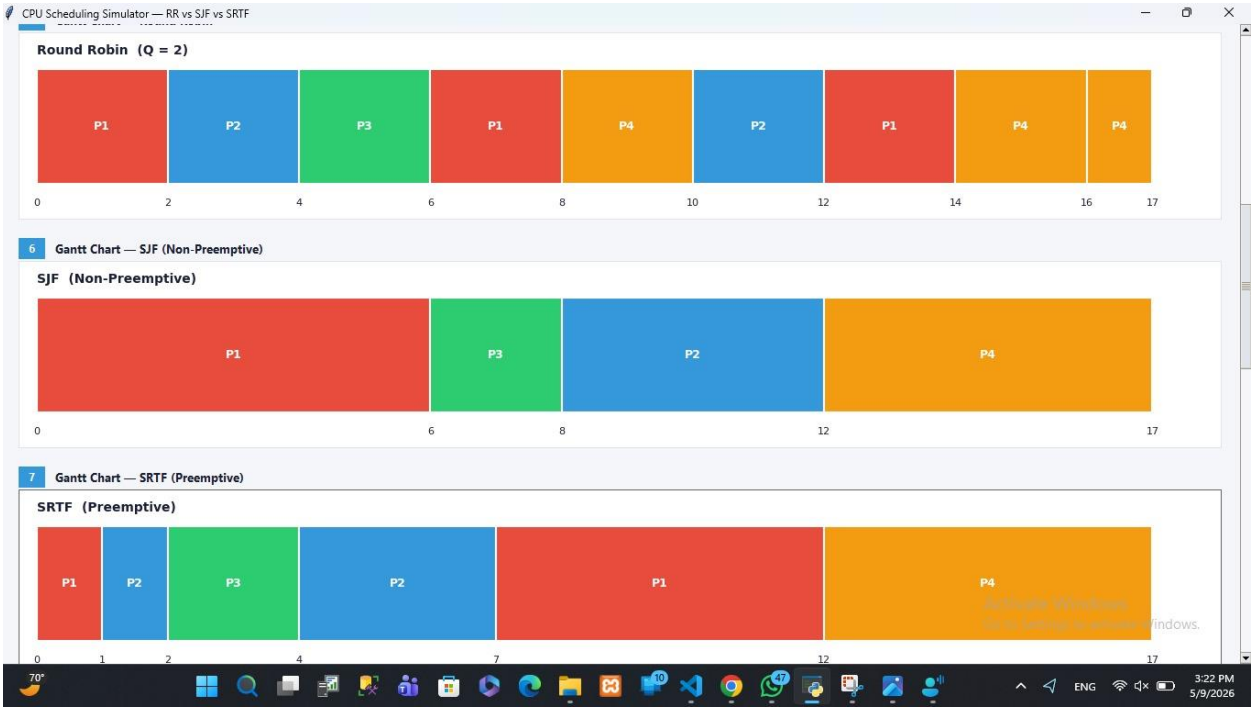
Scenario A:


CPU Scheduling Simulator — RR vs SJF vs SRTF
CPU Scheduling Simulator
Round Robin · SJF (Non-Preemptive) · SRTF (Preemptive)
Sections 1 & 2 — Input Panel + Quantum
PID
Arrival Time
Burst Time
+ Add Process
Time Quantum (Round Robin)
2
Must be a positive integer (> 0)
3 Process Table
PID | Arrival Time | Burst Time
P1 | 0 | 6
P2 | 1 | 4
P3 | 2 | 2
P4 | 3 | 5
Remove Selected
Run All Algorithms
Clear All
4 Ready Queue — Round Robin
Execution order (Q = 2):
P1 0→2
P2 2→4
P3 4→6
P1 6→8
P4 8→10
P2 10→12
P1 12→14
P4 14→16
P4 16→17
Activate Windows
Go to Settings to activate Windows.
Search
ENG
2:44 PM
5/9/2026


CPU Scheduling Simulator — RR vs SJF vs SRTF
Round Robin (Q = 2)
P1 P2 P3 P1 P4 P2 P1 P4 P4
0 2 4 6 8 10 12 14 16 17
6 Gantt Chart — SJF (Non-Preemptive)
SJF (Non-Preemptive)
P1 P3 P2 P4
0 6 8 12 17
7 Gantt Chart — SRTF (Preemptive)
SRTF (Preemptive)
P1 P2 P3 P2 P1 P4
0 1 2 4 7 12 17
ENG
3:22 PM
5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	6	0	14	0	14	8
P2	1	4	2	12	1	11	7
P3	2	2	4	6	2	4	2
P4	3	5	8	17	5	14	9
AVG	—	—	—	—	**2.00**	**10.75**	**6.50**

Round Robin (Q=2) → Avg WT: 6.50 Avg TAT: 10.75 Avg RT: 2.00

9 Results Table — SJF (Non-Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	6	0	6	0	6	0
P2	1	4	8	12	7	11	7
P3	2	2	6	8	4	6	4
P4	3	5	12	17	9	14	9
AVG	—	—	—	—	**5.00**	**9.25**	**5.00**

SJF → Avg WT: 5.00 Avg TAT: 9.25 Avg RT: 5.00

10 Results Table — SRTF (Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	6	0	12	0	12	6
P2	1	4	1	7	0	6	2
P3	2	2	2	4	0	2	0
P4	3	5	12	17	9	14	9

Activate Windows
Go to Settings to activate Windows.

70° ENG 3:23 PM 5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

10 Results Table — SRTF (Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	6	0	12	0	12	6
P2	1	4	1	7	0	6	2
P3	2	2	2	4	0	2	0
P4	3	5	12	17	9	14	9
AVG	—	—	—	—	**2.25**	**8.50**	**4.25**

SRTF → Avg WT: 4.25 Avg TAT: 8.50 Avg RT: 2.25

11 Comparison Summary

Metric	Round Robin (Q=2)	SJF	SRTF	Winner ✓
Avg Waiting Time	6.50	5.00	4.25	**SRTF**
Avg Turnaround Time	10.75	9.25	8.50	**SRTF**
Avg Response Time	2.00	5.00	2.25	**Round Robin (Q=2)**

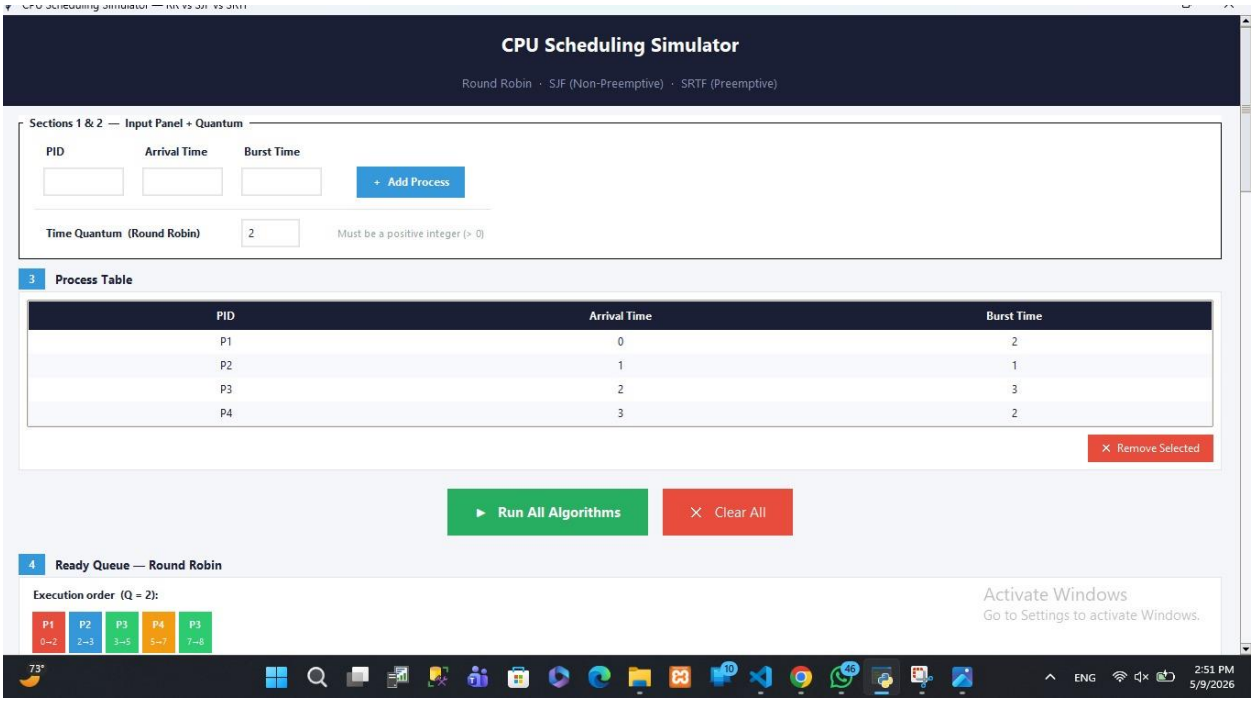
12 Final Conclusion

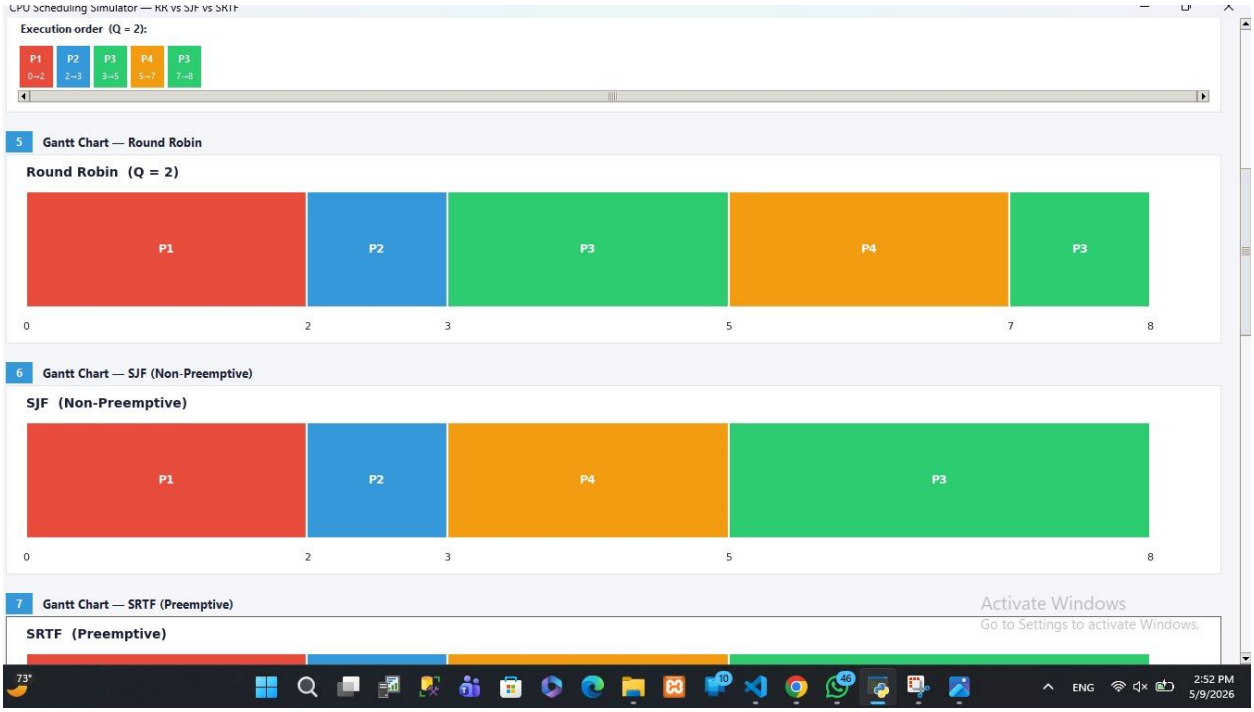
```
            scenarios; long jobs may wait if short jobs keep arriving.

  • Preemption:  SRTF (preemptive SJF) generally achieves the lowest
                 waiting and turnaround times but has higher context-switch
                 overhead and can starve long processes.

RECOMMENDATION
──────────────────────────────────────────────────────────────────────
  → Round Robin  — interactive / time-sharing systems (fairness matters).
```

Activate Windows
Go to Settings to activate Windows.

73° ENG 2:48 PM 5/9/2026

Scenario B:


CPU Scheduling Simulator
Round Robin · SJF (Non-Preemptive) · SRTF (Preemptive)
Sections 1 & 2 — Input Panel + Quantum
PID
Arrival Time
Burst Time
+ Add Process
Time Quantum (Round Robin)
2
Must be a positive integer (> 0)
3 Process Table
PID Arrival Time Burst Time
P1 0 2
P2 1 1
P3 2 3
P4 3 2
Remove Selected
Run All Algorithms
Clear All
4 Ready Queue — Round Robin
Execution order (Q = 2):
P1 0→2
P2 2→3
P3 3→5
P4 5→7
P3 7→8
Activate Windows
Go to Settings to activate Windows.
ENG
2:51 PM
5/9/2026


Execution order (Q = 2):
P1 0→2
P2 2→3
P3 3→5
P4 5→7
P3 7→8
5 Gantt Chart — Round Robin
Round Robin (Q = 2)
P1
P2
P3
P4
P3
0
2
3
5
7
8
6 Gantt Chart — SJF (Non-Preemptive)
SJF (Non-Preemptive)
P1
P2
P4
P3
0
2
3
5
8
7 Gantt Chart — SRTF (Preemptive)
SRTF (Preemptive)
Activate Windows
Go to Settings to activate Windows.
ENG
2:52 PM
5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

7 Gantt Chart — SRTF (Preemptive)

8 Results Table — Round Robin

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	2	0	2	0	2	0
P2	1	1	2	3	1	2	1
P3	2	3	3	8	1	6	3
P4	3	2	5	7	2	4	2
AVG	—	—	—	—	**1.00**	**3.50**	**1.50**

Round Robin (Q=2) → Avg WT: 1.50 Avg TAT: 3.50 Avg RT: 1.00

9 Results Table — SJF (Non-Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	2	0	2	0	2	0
P2	1	1	2	3	1	2	1
P3	2	3	5	8	3	6	3
P4	3	2	3	5	0	2	0

Activate Windows
Go to Settings to activate Windows.

73° ENG 2:52 PM 5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

10 Results Table — SRTF (Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	2	0	2	0	2	0
P2	1	1	2	3	1	2	1
P3	2	3	5	8	3	6	3
P4	3	2	3	5	0	2	0
AVG	—	—	—	—	**1.00**	**3.00**	**1.00**

SRTF → Avg WT: 1.00 Avg TAT: 3.00 Avg RT: 1.00

11 Comparison Summary

Metric	Round Robin (Q=2)	SJF	SRTF	Winner ✓
Avg Waiting Time	1.50	1.00	1.00	**SJF**
Avg Turnaround Time	3.50	3.00	3.00	**SJF**
Avg Response Time	1.00	1.00	1.00	**Round Robin (Q=2)**

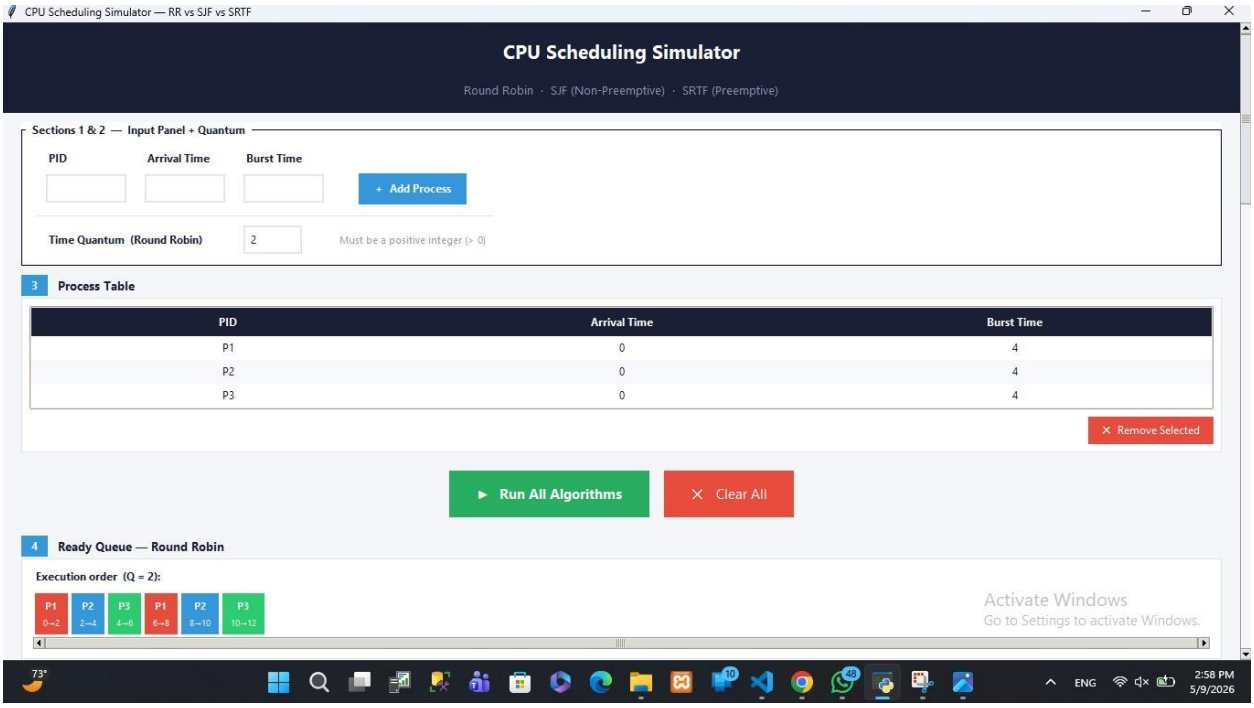
12 Final Conclusion

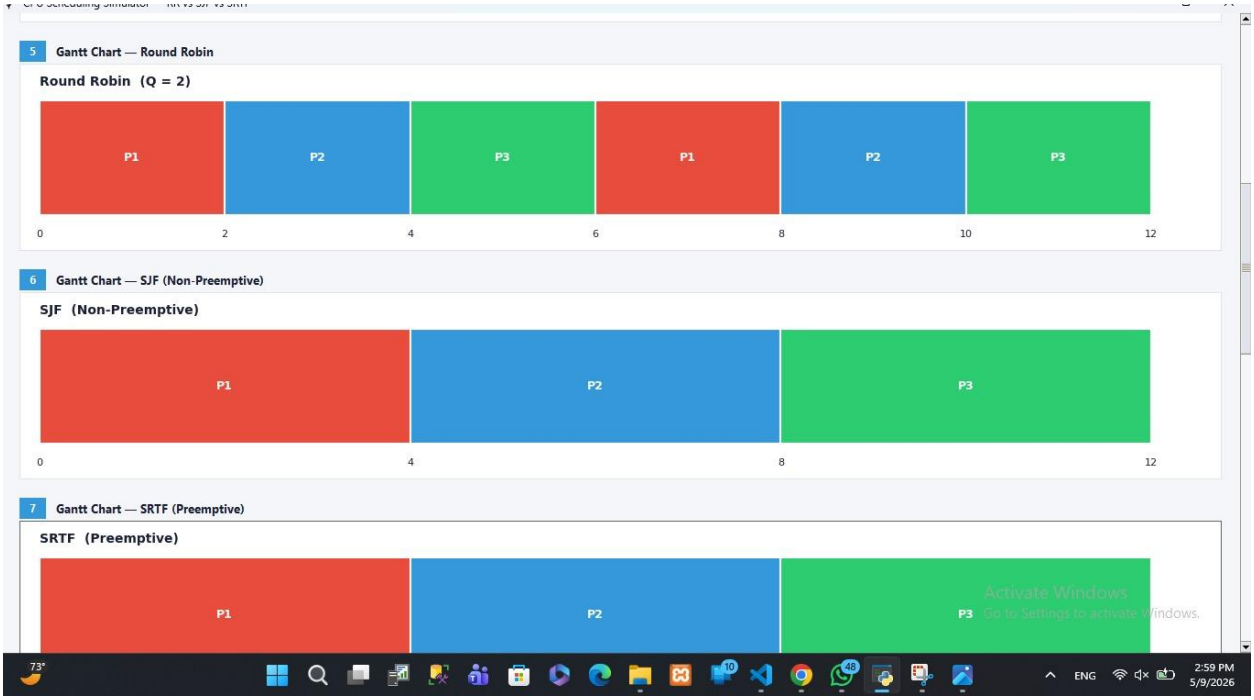
```
                scenarios; long jobs may wait if short jobs keep arriving.

• Preemption:   SRTF (preemptive SJF) generally achieves the lowest
                waiting and turnaround times but has higher context-switch
                overhead and can starve long processes.

RECOMMENDATION
──────────────────────────────────────────────────────────────────────
→ Round Robin  — interactive / time-sharing systems (fairness matters).
```

Activate Windows
Go to Settings to activate Windows.

73° ENG 2:53 PM 5/9/2026

Scenario C :


CPU Scheduling Simulator — RR vs SJF vs SRTF
CPU Scheduling Simulator
Round Robin · SJF (Non-Preemptive) · SRTF (Preemptive)
Sections 1 & 2 — Input Panel + Quantum
PID
Arrival Time
Burst Time
+ Add Process
Time Quantum (Round Robin)
2
Must be a positive integer (> 0)
3 Process Table
PID | Arrival Time | Burst Time
P1 | 0 | 4
P2 | 0 | 4
P3 | 0 | 4
✕ Remove Selected
▶ Run All Algorithms
✕ Clear All
4 Ready Queue — Round Robin
Execution order (Q = 2):
P1 0→2
P2 2→4
P3 4→6
P1 6→8
P2 8→10
P3 10→12
Activate Windows
Go to Settings to activate Windows.
ENG
2:58 PM
5/9/2026


5 Gantt Chart — Round Robin
Round Robin (Q = 2)
P1
P2
P3
P1
P2
P3
0
2
4
6
8
10
12
6 Gantt Chart — SJF (Non-Preemptive)
SJF (Non-Preemptive)
P1
P2
P3
0
4
8
12
7 Gantt Chart — SRTF (Preemptive)
SRTF (Preemptive)
P1
P2
P3
Activate Windows
Go to Settings to activate Windows.
ENG
2:59 PM
5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

SRTF (Preemptive)

8 Results Table — Round Robin

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	4	0	8	0	8	4
P2	0	4	2	10	2	10	6
P3	0	4	4	12	4	12	8
AVG	—	—	—	—	**2.00**	**10.00**	**6.00**

Round Robin (Q=2) → Avg WT: 6.00 Avg TAT: 10.00 Avg RT: 2.00

9 Results Table — SJF (Non-Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	4	0	4	0	4	0
P2	0	4	4	8	4	8	4
P3	0	4	8	12	8	12	8
AVG	—	—	—	—	**4.00**	**8.00**	**4.00**

Activate Windows
Go to Settings to activate Windows.

SJF → Avg WT: 4.00 Avg TAT: 8.00 Avg RT: 4.00

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CPU Scheduling Simulator — RR vs SJF vs SRTF

10 Results Table — SRTF (Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	4	0	4	0	4	0
P2	0	4	4	8	4	8	4
P3	0	4	8	12	8	12	8
AVG	—	—	—	—	**4.00**	**8.00**	**4.00**

SRTF → Avg WT: 4.00 Avg TAT: 8.00 Avg RT: 4.00

11 Comparison Summary

Metric	Round Robin (Q=2)	SJF	SRTF	Winner ✓
Avg Waiting Time	6.00	4.00	4.00	**SJF**
Avg Turnaround Time	10.00	8.00	8.00	**SJF**
Avg Response Time	2.00	4.00	4.00	**Round Robin (Q=2)**

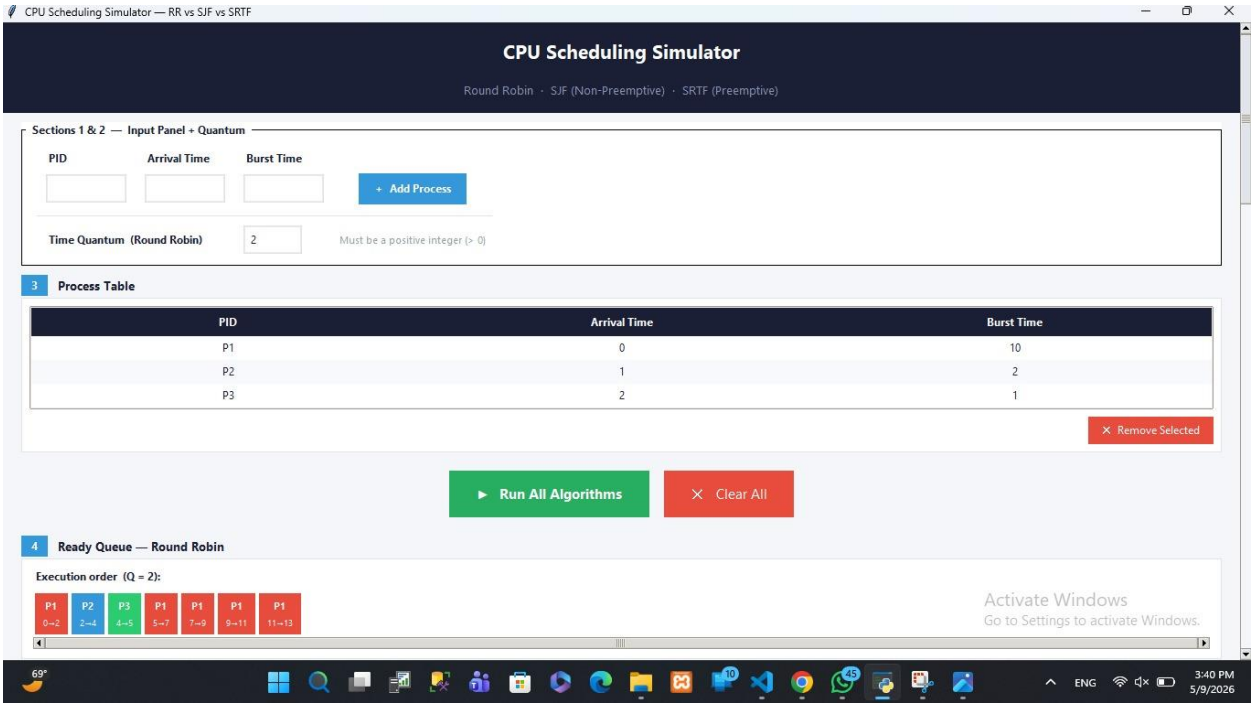
12 Final Conclusion

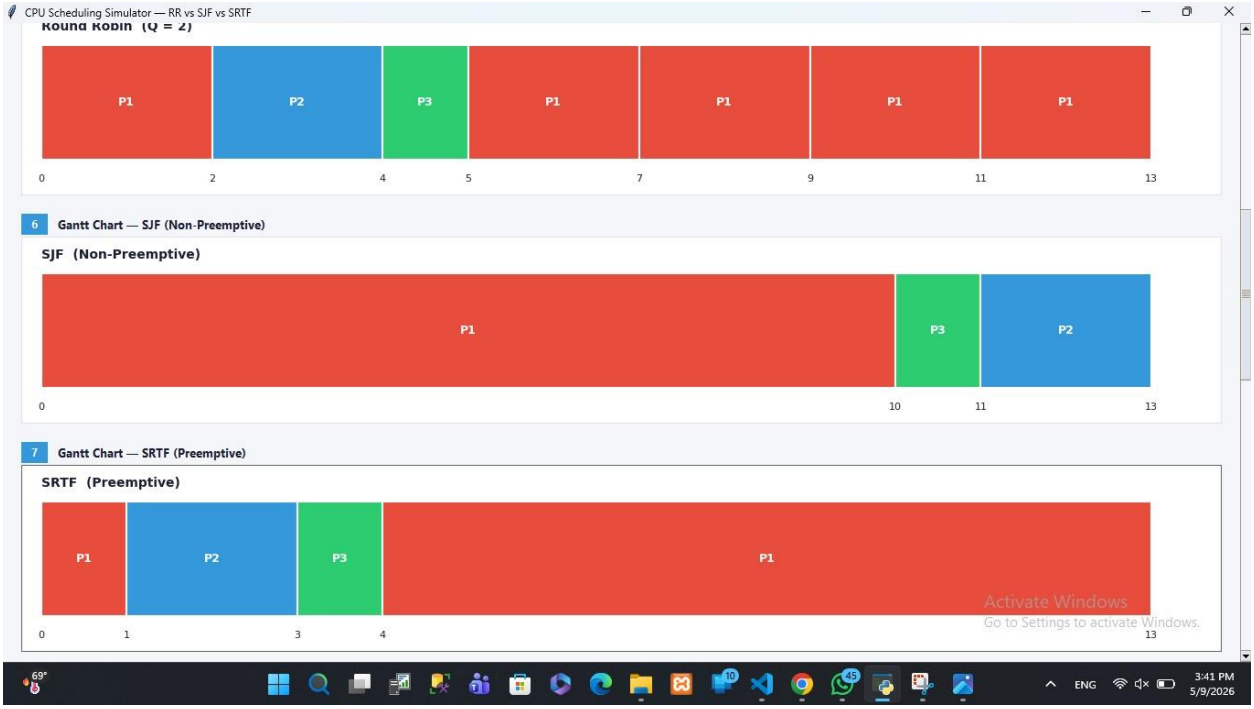
```
PERFORMANCE SUMMARY

Avg Waiting Time    →  RR=6.00  SJF=4.00  SRTF=4.00   Best: Tie (SJF / SRTF)
Avg Turnaround Time →  RR=10.00  SJF=8.00  SRTF=8.00   Best: Tie (SJF / SRTF)
Avg Response Time   →  RR=2.00  SJF=4.00  SRTF=4.00   Best: Round Robin (Q=2)

ANALYSIS
```

Activate Windows
Go to Settings to activate Windows.

73° ENG 3:00 PM 5/9/2026

Scenario D:


CPU Scheduling Simulator — RR vs SJF vs SRTF
CPU Scheduling Simulator
Round Robin · SJF (Non-Preemptive) · SRTF (Preemptive)
Sections 1 & 2 — Input Panel + Quantum
PID
Arrival Time
Burst Time
+ Add Process
Time Quantum (Round Robin)
2
Must be a positive integer (> 0)
3 Process Table
PID Arrival Time Burst Time
P1 0 10
P2 1 2
P3 2 1
✕ Remove Selected
▶ Run All Algorithms
✕ Clear All
4 Ready Queue — Round Robin
Execution order (Q = 2):
P1 0→2
P2 2→4
P3 4→5
P1 5→7
P1 7→9
P1 9→11
P1 11→13
Activate Windows
Go to Settings to activate Windows.
ENG
3:40 PM
5/9/2026


CPU Scheduling Simulator — RR vs SJF vs SRTF
Round Robin (Q = 2)
P1 P2 P3 P1 P1 P1 P1
0 2 4 5 7 9 11 13
6 Gantt Chart — SJF (Non-Preemptive)
SJF (Non-Preemptive)
P1 P3 P2
0 10 11 13
7 Gantt Chart — SRTF (Preemptive)
SRTF (Preemptive)
P1 P2 P3 P1
0 1 3 4 13
Activate Windows
Go to Settings to activate Windows.
ENG
3:41 PM
5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

8 Results Table — Round Robin

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	10	0	13	0	13	3
P2	1	2	2	4	1	3	1
P3	2	1	4	5	2	3	2
AVG	—	—	—	—	**1.00**	**6.33**	**2.00**

Round Robin (Q=2) → Avg WT: 2.00 Avg TAT: 6.33 Avg RT: 1.00

9 Results Table — SJF (Non-Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	10	0	10	0	10	0
P2	1	2	11	13	10	12	10
P3	2	1	10	11	8	9	8
AVG	—	—	—	—	**6.00**	**10.33**	**6.00**

SJF → Avg WT: 6.00 Avg TAT: 10.33 Avg RT: 6.00

10 Results Table — SRTF (Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	10	0	13	0	13	3
P2	1	2	1	3	0	2	0
P3	2	1	3	4	1	2	1
AVG	—	—	—	—	**0.33**	**5.67**	**1.33**

Activate Windows
Go to Settings to activate Windows.

69° ENG 3:41 PM 5/9/2026

CPU Scheduling Simulator — RR vs SJF vs SRTF

10 Results Table — SRTF (Preemptive)

PID	Arrival	Burst	Start	Finish	RT	TAT	WT
P1	0	10	0	13	0	13	3
P2	1	2	1	3	0	2	0
P3	2	1	3	4	1	2	1
AVG	—	—	—	—	**0.33**	**5.67**	**1.33**

SRTF → Avg WT: 1.33 Avg TAT: 5.67 Avg RT: 0.33

11 Comparison Summary

Metric	Round Robin (Q=2)	SJF	SRTF	Winner ✓
Avg Waiting Time	2.00	6.00	1.33	**SRTF**
Avg Turnaround Time	6.33	10.33	5.67	**SRTF**
Avg Response Time	1.00	6.00	0.33	**SRTF**

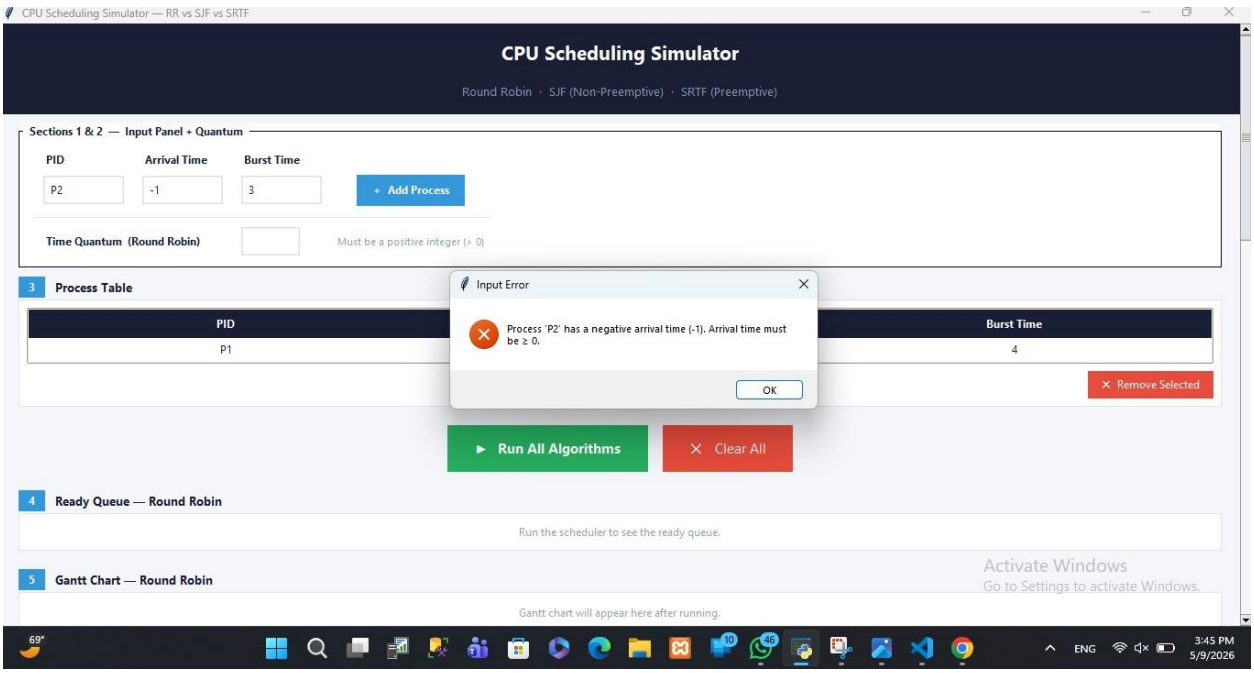
12 Final Conclusion

```
PERFORMANCE SUMMARY

Avg Waiting Time    → RR=2.00  SJF=6.00  SRTF=1.33   Best: SRTF
Avg Turnaround Time → RR=6.33  SJF=10.33  SRTF=5.67   Best: SRTF
Avg Response Time   → RR=1.00  SJF=6.00  SRTF=0.33   Best: SRTF

ANALYSIS
```

Activate Windows
Go to Settings to activate Windows.

69° ENG 3:42 PM 5/9/2026

Scenario E:


CPU Scheduling Simulator — RR vs SJF vs SRTF
CPU Scheduling Simulator
Round Robin · SJF (Non-Preemptive) · SRTF (Preemptive)
Sections 1 & 2 — Input Panel + Quantum
PID
Arrival Time
Burst Time
P2
-1
3
+ Add Process
Time Quantum (Round Robin)
Must be a positive integer (> 0)
3 Process Table
PID
Burst Time
P1
4
Input Error
Process 'P2' has a negative arrival time (-1). Arrival time must be ≥ 0.
OK
Remove Selected
Run All Algorithms
Clear All
4 Ready Queue — Round Robin
Run the scheduler to see the ready queue.
5 Gantt Chart — Round Robin
Gantt chart will appear here after running.
Activate Windows
Go to Settings to activate Windows.
ENG
3:45 PM
5/9/2026